IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-017

1. Topic (12 words max)

Secure Microfinance and Welfare Management Platform Using Blockchain and AI

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	R26-IT-017
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

In Sri Lanka, microfinance and welfare programs such as Samurdhi, Aswasuma, and community-based microfinance schemes play a crucial role in poverty alleviation, rural development, and women's economic empowerment. Research evidence shows that microfinance has positively contributed to household income growth, poverty reduction, and social empowerment, particularly among low-income and rural communities. Despite these benefits, the current management of microfinance and welfare services at the Grama Niladhari (GN) Division level faces significant operational and governance challenges.

A key problem is the heavy reliance on manual, fragmented, and paper-based processes for beneficiary registration, eligibility verification, loan approval, and repayment monitoring. These practices lead to data duplication, delayed service delivery, biased decision-making, and limited transparency. GN and Samurdhi officers face high administrative workloads, while microfinance institutions struggle with inaccurate risk assessment and increasing loan default rates. As a result, deserving beneficiaries may be excluded, while fraudulent or duplicate claims may go undetected.

The impact of this problem is multifaceted. Socially, inefficient welfare distribution reduces trust in public institutions and weakens empowerment outcomes, particularly for women and vulnerable groups. Economically, poor monitoring and lack of predictive mechanisms increase financial risks and reduce the sustainability of microfinance programs.

Technically, the absence of an integrated digital platform results in data silos and low interoperability across institutions. Ethically and legally, manual systems increase opportunities for corruption, data manipulation, and privacy violations.
Although Sri Lanka has initiated digital government reforms, studies indicate that secure and intelligent digital systems are still limited at the local governance level. Therefore, the primary research problem is to develop a secure, transparent, and intelligent digital system to improve microfinance and welfare management at the GN Division level. Addressing this problem can strengthen governance, reduce inequalities, and promote inclusive and sustainable economic development in Sri Lanka.

References

[1] H. M. W. A. Herath, L. H. P. Gunaratne, and N. Sanderatne (2015) *Impact of Microfinance on Poverty Reduction: Evidence from Sri Lanka*, University of Peradeniya.

[2] M. A. M. I. Perera and W. M. S. M. Wijekoon (2019) *Role of Microfinance in Empowering Poor Rural Households in Sri Lanka*, Sri Lankan Journal of Business Economics.

[3] V. Perera (2021) *Women Empowerment Through Microfinance: An Empirical Study on Ampara District*, International Conference on Business Management.

[4] N. Kannangara and C. Liyanage (2023) *Effectiveness of Microfinance Services on Poverty Alleviation in Sri Lanka*, South Asian Journal of Finance.

[5] World Bank (2020) *Financial Consumer Protection and Microfinance*, World Bank Publications.

[6] A. Mudalige (2021) *Adoption of Digital Government Technologies by Government Organizations in Sri Lanka*, SLIIT Research Data Archive.

[7] United Nations (2015) *Transforming Our World: The 2030 Agenda for Sustainable Development*.

6. Brief description of Existing Research and Systems (300 words max)

In recent years, blockchain technology has gained attention in the microfinance domain as a means of improving transparency, security, and trust. Several research studies propose blockchain-based microfinance systems where loan disbursement, repayment transactions, and borrower credit histories are recorded on decentralized ledgers. These systems mainly aim to reduce fraud, eliminate intermediaries, and ensure transparent financial transactions between lenders and borrowers [1], [2]. Some studies further integrate smart contracts to automate loan approval processes, repayment schedules, and penalty enforcement, thereby reducing manual intervention and operational costs [3].

Some existing research integrates smart contracts into blockchain-based microfinance systems to automate loan approval processes, repayment schedules, and penalty enforcement. This automation reduces manual intervention and improves operational efficiency in microfinance institutions [3]. In addition, blockchain has been combined with digital identity and Know Your Customer (KYC) mechanisms to prevent duplicate borrower registrations and identity fraud in micro-lending environments [4].

Alongside microfinance, several studies focus on digital welfare management systems that aim to improve the efficiency of welfare distribution and reduce administrative delays. These systems typically use centralized databases and management information systems (MIS) to store beneficiary data and track welfare disbursements [5]. While some systems introduce audit logs, they still rely on trusted central authorities and remain vulnerable to data manipulation and lack of transparency across stakeholders.

Artificial intelligence (AI) has also been applied in microfinance research to predict loan repayment behavior, assess credit risk, and identify high-risk borrowers using historical financial data [6], [7]. However, these AI-based systems usually operate independently of blockchain-based data governance mechanisms. Similarly, natural language processing (NLP) has been explored in limited contexts such as chatbots and automated form handling, but multilingual support for low-resource languages such as Sinhala and Tamil remains minimal [8].

Overall, existing research and systems address microfinance management, welfare distribution, AI-based prediction, and digital communication as separate solutions. There is a noticeable lack of an integrated platform that combines blockchain-based data integrity, government-level verification, intelligent prediction, and multilingual citizen interaction within a local administrative context.

References

[1] A. Tapscott and D. Tapscott, *Blockchain Revolution: How the Technology Behind Bitcoin Is Changing Money, Business, and the World*, Penguin Random House, 2016.

[2] N. Kshetri, “Blockchain’s roles in strengthening cybersecurity and protecting privacy,” *Telecommunications Policy*, vol. 41, no. 10, pp. 1027–1038, 2017.

[3] M. Swan, *Blockchain: Blueprint for a New Economy*, O’Reilly Media, 2015.

[4] S. Nakamoto, “Bitcoin: A peer-to-peer electronic cash system,” 2008.

[5] United Nations Development Programme (UNDP), *Digital Governance and Welfare Systems*, UNDP Publications, 2020.

[6] R. M. Bellotti et al., “Data mining methods for credit risk analysis in microfinance,” *Computational Economics*, vol. 47, no. 1, pp. 1–24, 2016.

[7] S. Moro, P. Cortez, and P. Rita, “A data-driven approach to predict the success of bank telemarketing,” *Decision Support Systems*, vol. 62, pp. 22–31, 2014.

[8] R. Joshi, P. Kumar, and A. Mishra, “Multilingual chatbot systems using natural language processing,” *International Journal of Computer Applications*, vol. 176, no. 39, pp. 12–18, 2020.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution, is a blockchain-integrated microfinance and welfare management system designed to address the limitations identified in existing research and systems discussed in Section 6. Current studies largely treat microfinance management, welfare distribution, AI-based prediction, and digital communication as independent solutions. There is a significant gap in the availability of a unified platform that combines secure data governance, government-level verification, intelligent decision support, and inclusive citizen interaction within a localized administrative context. The system platform aims to bridge this gap by integrating blockchain technology, smart contracts, artificial intelligence (AI), and multilingual natural language processing (NLP) into a single system.

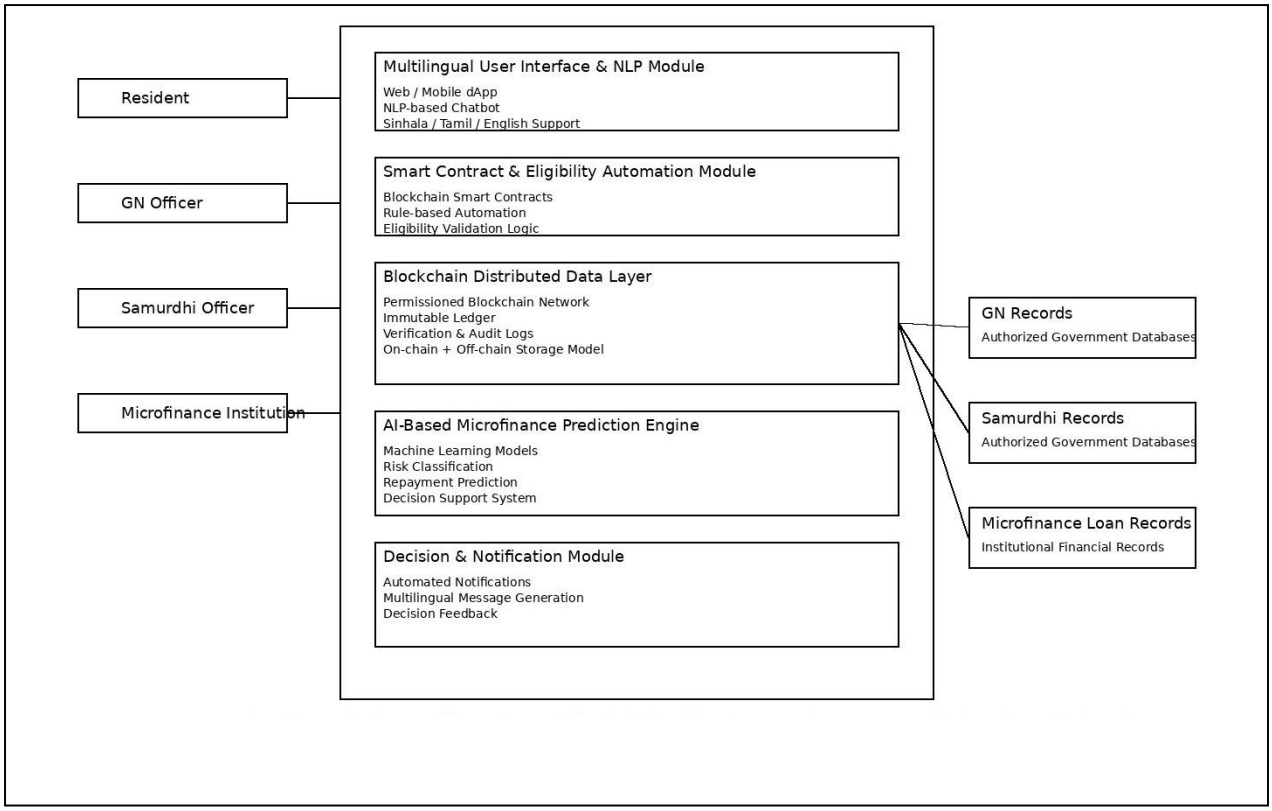
Unlike existing blockchain-based microfinance systems that primarily focus on transparent financial transactions, the proposed system extends blockchain usage to include welfare records, verification logs, and eligibility decisions, ensuring data integrity and accountability across multiple stakeholders. The incorporation of smart contracts automates eligibility validation and rule enforcement, reducing manual intervention and minimizing fraud. Furthermore, while prior AI-based credit risk prediction models operate independently of secure data governance mechanisms, this system tightly integrates AI prediction with blockchain-verified data, enhancing the reliability of risk assessments and loan recommendations. The inclusion of a multilingual NLP-based user interface addresses the lack of accessibility for low-resource languages such as Sinhala and Tamil, enabling effective citizen interaction at the grassroots level.

System Perspective:

Input:
The system accepts inputs such as welfare or microfinance applications submitted by residents through a multilingual web or mobile interface. Additional inputs include verification data from GN officers and Samurdhi officers, as well as historical welfare and microfinance records obtained from authorized institutions.

Process:
Submitted applications are first processed through the user interface and NLP module, which enables multilingual communication. The data is then validated using smart contracts, which automatically check eligibility criteria and verification rules. Verified information is securely stored on a blockchain-based distributed ledger, ensuring transparency and immutability. Subsequently, the AI-based microfinance prediction engine analyzes blockchain-verified data to assess risk levels, predict repayment behavior, and generate loan or welfare recommendations.

Output:
The system produces outputs such as approval or rejection decisions, risk scores, and recommended repayment plans. These decisions are communicated to residents and relevant microfinance institutions through automated notifications in the user's preferred language.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

01.BlockChain integrated microfinance and welfare management platform

Domain Expertise : Blockchain system for public- sector governance, Microfinance operations, welfare administration in Grama niladhari division ,decentralized application (dApp) development ,digital identity management

Knowledge : Understand blockchain architectures (hybrid and permissioned blockchains), smart contract development (eg : Ethereum /Hyperledger), decentralized data governance , Grama niladhari and samurdhi welfare workflows. Microfinance eligibility rules, and secure off-chain, on-chain data integration .

Data Availability : Resident demographic data, household income, and expenditure records, welfare eligibility data (samurdhi , uswasuma), microfinance loan records, repayment histories and verification logs from GN division and microfinance institution such as Sanasa. Data will be accessed through institutional collaboration , anonymized dataset or simulated dataset .

02.Smart Contract and dApp based eligibility Automation System

Domain Expertise : Decentralized finance (DeFi) concepts,rule based eligibility systems, welfare policy enforcement and digital public service automations.
Knowledge : Smart contract login design, eligibility rule modelling, fraud and duplicate claim detection , Blockchain based audit trails and role based access control for GN officers, samurdhi officers, residents, and microfinance institution.
Data Availability : eligible criteria datasets, welfare policy rules, historical approval/rejection records, repayment compliance data, and application verification logs collected from GN and samurdhi administrative processes.

03.AIbase Microfinance prediction engine

Domain Expertise *:* Financial risk modelling, microfinance analytics,applied machine learning , and socio- economic data analysis
Knowledge : Supervised machine learning(Classification and regression), feature engineering for financial data , bias detection and mitigation ,risk scoring models,and repayment behaviour analysis.
Data Availability : Historical microfinance loan dataset, repayment timelines, default records, income level data, welfare dependency indicators, and GN verified socio-economic profiles, where direct access is restricted ,ethically generated synthetic datasets based on real statistical distributions will be used.

04.NLP Powered multilingual communication and document automation

Domain Expertise : Multilingual NLP, conversational AI, Accessibility – focused system design, and digital public communication system
Knowledge : natural language processing for low – resource languages (Sinhala and Tamil), speech to text and text to speech system, intent detection , form data extraction, and chatbot development.
Data Availability : Multilingual text corpora (Sinhala, Tamil ,English) ,GN and welfare related documents and forms, notification templates, resident query logs, and publicly available language datasets supplemented with domain specific data collected during the research

9. Objectives and Novelty

Main Objective To design and develop a secure, transparent, and intelligent digital platform that integrates blockchain, smart contracts, artificial intelligence, and multilingual natural language processing to improve microfinance and welfare management at the Grama Niladhari Division level by ensuring accountable decision-making, accurate risk assessment, and inclusive citizen access.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22194626	Blockchain-based smart contract and dApp that automatically evaluates eligibility for microfinance loans and welfare programs using predefined rule-based criteria	Review existing microfinance systems and blockchain-based platforms. Define a limited set of eligibility criteria (income level, welfare history, repayment history, duplicate claims). Design and develop smart contracts for automated eligibility evaluation. Implement blockchain logging for transparent and auditable decisions. Develop a dApp interface for residents, GN/Samurdhi officers, and microfinance institutions. Integrate the dApp with smart contracts for real-time evaluation.	Automating eligibility decision-making, rather than focusing only on loan disbursement or repayment as seen in most existing microfinance dApps. Introduces rule-based smart contracts for eligibility evaluation, multi-stakeholder verification within a single decentralized platform, and immutable blockchain records of eligibility decisions.

		Test the system using sample data and analyze results.	
IT22145048	Blockchain-Based Distributed Data Layer for Microfinance & Welfare Governance	Create a permissioned blockchain architecture that can be administered at the GN division level. Determine which data elements should be kept off-chain and on-chain for privacy and scalability. Ensure that GN and Samurdhi officer verification and approval actions are securely recorded. Create safeguards against unauthorized record changes, duplicate welfare claims, and data manipulation. Make it possible for GN offices, welfare authorities, and microfinance organizations to access shared data.	Proof-of-Verification blockchain ledger that records government officer approvals as immutable governance events. Unlike existing blockchain microfinance systems that focus mainly on financial transactions, this component secures welfare eligibility decisions and human verification actions, creating accountability and trust at the GN-division level.
IT22111074	An AI-based microfinance prediction engine that analyzes GN-division-level socio-economic, welfare, and microfinance data to predict applicant risk levels, repayment	Collect and anonymize socio-economic and microfinance data from GN divisions, Samurdhi records, and microfinance institutions, including income, welfare history, loan details, and repayment behavior. Extract key indicators such as income-to-loan ratio, welfare dependency level, and repayment consistency, and preprocess data for model training.	Two-way decision support, providing risk insights to microfinance institutions and affordability guidance to residents. Integration of welfare and microfinance data for improved risk assessment and misuse detection. Bias-aware financial risk modeling, promoting ethical and transparent AI

	duration, and financial stability, while providing bias-aware, two-way decision support	Develop supervised machine learning models to classify applicants into low, medium, and high risk categories based on historical data. Use regression models to estimate repayment duration and predict financial stability trends. Evaluate models for bias and integrate predictions into the system as decision support insights for institutions and residents.	use in rural and public-sector financial decision-making.
IT22552310	NLP-Powered Multilingual Communication & Document Automation	Design a multilingual NLP architecture supporting Sinhala, Tamil, and English. Develop an AI chatbot to handle resident queries related to welfare, loans, and GN services. Implement natural language intent detection for domain-specific queries (microfinance, welfare, eligibility). Develop automated message generation for welfare notifications, loan reminders, and deadlines.	Integration of low-resource language NLP (Sinhala & Tamil) within a GN Division–level microfinance and welfare system. Unified chatbot-based communication replacing traditional manual notices and in-person inquiries.

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10. <u>This part is to be filled by the Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	x	No	

b) The proposed topic exhibits novelty.

Yes	x	No	

c) The student group can successfully execute the proposed project.

Yes	x	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	x	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Dharshana	Kasthurirathna	
Co-Supervisor	Ms.	Chathurya	Kumarapperuma	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				